

MA-105 Tutorial-6

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1. Show that $\{(x_1, x_2, x_3) \mid x_1 + x_2 + x_3 > 0\}$ is open.
2. Let P, Q, R be three non collinear points in $\mathbb{R}^2$. Define precisely the closed triangle determined by P, Q, R . Is it convex? Is the open triangle determined by P, Q, R convex?
3. Let $f : \mathbb{I} \rightarrow \mathbb{R}$ be convex and $g : \mathbb{I} \rightarrow \mathbb{R}$ be concave, where $\mathbb{I}$ is an open interval in $\mathbb{R}$.

Assume $g(x) > f(x) \forall x \in \mathbb{I}$.

Show that the regions $G_1 = \{(x, y) \in \mathbb{R}^2 \mid x \in \mathbb{I}, y < g(x)\}$, $G_2 = \{(x, y) \in \mathbb{R}^2 \mid x \in \mathbb{I}, y > f(x)\}$ and $G = \{(x, y) \in \mathbb{R}^2 \mid x \in \mathbb{I}, f(x) < y < g(x)\}$ all are convex.

4. Let

$$f(x, y) = \begin{cases} \frac{x^2 y}{x^4 + y^2} & (x, y) \neq (0, 0) \\ 0 & (x, y) = (0, 0) \end{cases}$$

Show that $f(x, y)$ is not continuous at the origin by taking two different sequences of vectors v_n approaching $(0, 0)$ and showing that $f(v_n)$ approach different vectors in both cases.

5. Sketch the following loci in $\mathbb{R}^3$:

- $z = x^2 + y^2$
- $z = x^2 - y^2$
- $z^2 = x^2 + y^2$
- $z^2 = x^2 + y^2 + \alpha^2$ ($\alpha \neq 0$)
- $z^2 = x^2 + y^2 - \alpha^2$ ($\alpha \neq 0$)

6. Sketch the following loci in $\mathbb{R}^2$:

- $\frac{y^2}{2} - \frac{g}{l} \sin x = E$, where E is constant
- $((x - a)^2 + y^2)((x + a)^2 + y^2) = C$, where C is a constant

Find the points of maxima, minima and saddle point of all these curves.